BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 066 · Array Traversal and Updates

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a one-dimensional array?

Q2. Valid ages are 13–18. Which is normal test data?
A. 12
B. 15
C. 19
D. 'twelve'

Q3. “Any device connected to the Internet is automatically an embedded system.” Correct this. Explain why.

Q4. A pupil enters a domain name. “DNS gives the same MAC address to every visitor.” Correct this. How does DNS help find the website?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Traversal — A loop visits every array element using a valid index.

Accumulator — A total variable stores the running sum and is updated for each value.

FOR iteration — A counted loop can process indices zero through length minus one.

Average — Total divided by number of elements gives the mean.

Syntax — The rules governing valid programming statements.

Test data — An input used to check expected results.

New learning

Traversal: A loop visits every array element using a valid index. Apply this in a competition points.

Accumulator: A total variable stores the running sum and is updated for each value. Apply this in a four temperatures.

FOR iteration: A counted loop can process indices zero through length minus one. Apply this in a café takings.

Average: Total divided by number of elements gives the mean. Apply this in a weekly marks.

Worked example

A competition points is being developed. Explain how traversal and accumulator are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Traversal [1]; A loop visits every array element using a valid index. [1].

3. Accumulator [1]; A total variable stores the running sum and is updated for each value. [1].

4. Two linked points for each concept.

We Do · Practise with guidance

Guided 1

A four temperatures uses traversal. Explain how it helps. [2 marks]

Worked method: Traversal [1]; A loop visits every array element using a valid index. [1].

Guided 2

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Explain accumulator in a café takings. [2 marks]

Worked method: Accumulator [1]; A total variable stores the running sum and is updated for each value. [1].

Guided 3

A weekly marks uses for iteration and average. Explain both. [4 marks]

Worked method: FOR iteration [1]; A counted loop can process indices zero through length minus one. [1]. Average [1]; Total divided by number of elements gives the mean. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the line that adds each file size to the total.

```
array imageMB = [2,5,3]
total = 0
for i = 0 to 2
    // add imageMB[i]
next i
print(total)
```

write

Question 1

State TWO relevant techniques from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

Explain how accumulator and for iteration help with a fitness step counts. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A programmer designs a lunch orders. Explain why traversal, for iteration and average matter. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A loop visits every array element using a valid index.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A fixed-size indexed collection of related elements; OCR ERL indexing begins at zero.

Q2. B — 15

Q3. It is part of a heating system, with the specific task of controlling temperature.

Q4. DNS servers look up the IP address for a domain name. Access needs the relevant server and network services to work.

I Do / We Do pages · Guided practice

Q1. Traversal [1]; A loop visits every array element using a valid index. [1].

Q2. Accumulator [1]; A total variable stores the running sum and is updated for each value. [1].

Q3. FOR iteration [1]; A counted loop can process indices zero through length minus one. [1]. Average [1]; Total divided by number of elements gives the mean. [1].

You Do pages · Code First

```
array imageMB = [2,5,3]
total = 0
for i = 0 to 2
    total = total + imageMB[i]
next i
print(total)
```

Expected result: 10

You Do pages · Independent questions

Q1. Traversal [1]; FOR iteration [1].

Q2. Accumulator [1]; A total variable stores the running sum and is updated for each value. [1]. FOR iteration [1]; A counted loop can process indices zero through length minus one. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Traversal [1]; A loop visits every array element using a valid index. [1]. FOR iteration [1]; A counted loop can process indices zero through length minus one. [1]. Average [1]; Total divided by number of elements gives the mean. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: